

Writing A Report: InDaS 2 Edition

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1 Industrial Data Science 2 Specifics

Requirements for the report in InDaS 2:

- You are required to write individual reports.
- You may share figures and tables within your group. However, we expect you to be able to reproduce (in principle) and understand them!
- Exceeding the page limit results in a deduction of points. While figures and tables count towards this limit, table of contents, table of figures or similar do not.
- The (optional) appendix is not graded. You may use it for parameter tables or auxiliary information that is not explicitly discussed within the report.
- Describe only the method you chose in the end, no need to explain what you didn't use.
- Aside from a basic, compact definition of the method, describe if and how you adopted the method to fit your task.
- Structure your report logically: No questions should be left open and no section should feel out of place.
- Pre-processing/Feature engineering is very important and forms the foundation for the method - don't forget to describe it.

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- Connect "statistics" and "the real world":
 - Motivate your method/approach: What is the goal of your task and how can you model it statistically? Why have you chosen your specific method? Which features are statistically significant? If they are not, is excluding them sound?
 - Describe your data: What kind of variables are there, how are they distributed, scaled, ...?
 - Interpret your result: Are the rates acceptable for your use-case? What does the method's success mean for the use-case? Also, you do not have to defend bad results; we grade *how* you approached (and documented) the problem, not the success of your solution.

Things to look out for/discuss:

- Did the method overfit? How can overfitting be averted? Or conversely, can the approach generalize to new data? Extremely high classification rates should make you suspicious. Think about methods of regularization.
- If we have a classification problem, are the classes balanced?
- Do not use simple train/test, use some form of validation (e.g., cross validation).
- Are there other ways to form meaningful features from the existing ones?
- Are the features/variables directly comparable? E.g., are they of the same unit of measurement and variable type? Transformations might be required.
- What are simplifying or complicating circumstances of our use-case? Methods taught in lectures usually do not account for specific situations.
 - For instance, in image recognition tasks, do we have to account for rotation/ translation of the object, changes of background or obfuscation?
 - Do we know the geometry of the object and can we use that knowledge?
 - Can we extract the relevant information from the image and apply our model on a feature space more expressive than the image space?
 - And does a specific approach retain the ability to generalize to unseen instances within our use-case?
- Analyze errors of your approach. Can we explain (individual) results? I.e., do we know why the method made a specific type of error?
- Be aware that (algorithm) "runtime" in data science is not necessarily a major concern, always consider your use-case!

- Do not describe irrelevant steps with no major impact. For instance, it is a minor detail that data might have been "converted from .csv to a binary format". Changes to their inherent properties (e.g., floating point precision) can be relevant, however!

Lastly, you may shift the focus on your report slightly to reflect your focus in your group's work. We do not require you to explicitly state your exact contribution. Be aware, however, that you should not completely omit important parts. If you are unsure on what to include, omit or expand, feel free to ask your supervisors!

1.0.1 Additional Task for CS Students

Usually, the additional task can be realized as a direct comparison with a second method (aside from the one you chose with your group). This entails that you give a compact definition first, before discussing the merits and demerits.

Generally, your use-case might suggest a different task than a straight method comparison. This can, for example, be an alternative set of feature transforms (which usually requires retraining the method) or a variation of the initial task your group was given. Feel free to ask and confirm your intentions with your CS supervisor.

2 Report Structure

While it is not mandatory to follow the structure below, it gives you an idea about relevant/important aspects of your report.

Note that it is generally not advisable to write *chronologically*, focus on the result instead. To this end, you can simplify earlier attempts or even omit them, if their discussion is not worthwhile. Be aware, however, that "failed" attempts can also contain noteworthy insights!

- **Introduction:**
 - Motivate the problem in general and for your group's use-case
 - Define the problem & your goal
 - Briefly outline your result and the report
- **Data**
 - Detailed description of the data: distribution, quality, quantity, ...
 - Feature selection/engineering & other pre-processing steps
- **Methods**
 - Definition
 - Analytical strategy

- **Results**
 - Presentation (text, tables, figures) and discussion of the results: What is shown here?
 - Interpretation of the results
- **Deployment** (can be merged with Discussion)
 - What has to be done to apply the method in the real world?
- *Computer Science Students Only*: **Comparison with 2nd Method**
 - Second method definition
 - Detailed Comparison
 - Interpretation: Pro's and con's of each method?
- **Discussion / Conclusion / Summary**
 - Generally a short summary of the result (very brief)
 - Critical examination / limitations of your work: Did you achieve the goal?
 - Outlook / possible areas of improvement

3 Addendum: Scientific writing

This section gives some very general ideas about scientific writing, which you can apply for your report, thesis or research papers:

3.1 Grice's Maxims of Conversation

Utilize Grice's 4 maxims for ever meaningful discourse:

- As informative as required, but not more informative than needed (maxim of quantity).
- Do not say anything you believe to be false or for which you do not have sufficient evidence (maxim of truth).
- Be relevant: Contribute information that is relevant to the topic at hand (maxim of relation).
- Be precise: Avoid ambiguity, be brief and provide the information in a useful order (maxim of clarity).

3.2 General advices

In general, you should write objectively, i.e., your work should convey independence, neutrality and impartiality:

- To give some examples:
Don't: "I have ...", "...in my opinion...", "You can see..."
Instead: Passive voice, indirect speech when referring to yourself ("The author has...")
- No absolute statements, such as "... this will always be the case".
- No anecdotes.
- No metaphors, such as "As useful as a chocolate teapot."
- Fairness (e.g. counter opinions, alternative approaches)
- No rhetorical questions.
- No exaggerated or polemic speech.
- Write gender neutral. You can use the gender neutral pronoun "they", e.g., "A user may perform Y. However, they may also do Y.". Your texts should also be free of reproductions of discrimination against minorities.
- Use always color-blind friendly palettes in your graphics.

Also, you should write understandable and comprehensible:

- Write vividly and descriptively, i.e. replace bulky nouns with active verbs.
- Use short sentences.
- Write grammatically correct.
- Define unknown and new terms, always explain abbreviations.
- Include examples/graphics to accompany larger or complicated bits of information.
- Avoid graphics/figures with trivial information: there is no need to make a bar-plot for just two values. You can mention them in a simple sentence.
- Don't use figures without discussing them in your text.
- All elements in a figure should be easily readable. That also applies to digital documents.

Last recommendation: Learn L^AT_EX! ☺

4 Addendum: Reading Scientific Articles

Beyond an application of suitable methods, *research* requires considering the state of the art, usually documented in scientific articles. In the following, you can find some (very general) tips on how to approach reading such articles.

Reading technique

- **Superficial reading:**
 - Just skim the article in order to get a basic insight.
 - Do not get hung up on minor details.
 - Have the courage to stop reading prematurely and move on to another article.
- **Intensive reading:**
 - Read the article in order to fully understand the presented methods, results, etc.
 - Take your time. Even for the best of us, understanding a scientific article can be a challenge.
 - Be prepared to take a deeper dive in the given references.
 - Take a mental break and revisit the article after a couple of days.
- **Reading in layers:** from *superficial* to *very intensive* reading. Have I already found out what I wanted?
- **Highlight:** Develop a color scheme, but refrain from excessive use.
- **Notes:** Document your own thoughts (you will thank yourself in the future) and important aspects in Word/L^AT_EX or by hand. Think about an reference management software (e.g. Zotero) to organize the scientific articles you work with.
- **Visualization:** Sketches, diagrams, etc. help to understand difficult connections/contexts.
- **Stay calm:** It is normal to feel stupid.
- **Courage to skip:** Things you already know can be skipped.

Critical reading

- Critical reading = "reflect on the content and quality of the article"
- Critical reading $\neq$ "pure error searching & contradicting"
- Do I like the approach? Why/Why not?

- Are the conclusions reasonable? Complete?
- Is the research coherent and comprehensible?
- Can one replicate the results with the given description?
- Remain skeptical, even if the authors are well known and the publisher is highly ranked. Which assumptions were made and is the method as widely applicable as it was (perhaps implicitly) suggested? Do the assumptions of the past still hold?
- Look up who cited the work. → Gives clues about the quality, importance and scope of the topic.
- Experience and a good knowledge base helps with the assessment. → The more you read, the more knowledge you can use for a critical analysis.

Be aware that some terms have different meanings, depending on context. While this may sound obvious, the differences are sometimes subtle. It often helps to search the background/state-of-the-art sections in the cited works of a paper for more explicit explanations/definitions. Additionally, many topics have so called survey papers, which explain several variants of the same concept and help you in understanding the concept more broadly.